Retail Sales Analysis Report

Sales performance, customer behavior, product performance and monthly trends

Dataset: Retail Sales Dataset (1,000 transactions)
Period covered: January 2023 - January 2024
Tools: Microsoft Excel (analysis and dashboard)

1. Executive Summary

This report analyses 1,000 retail transactions across three product categories (Beauty, Clothing and Electronics) to answer six business questions on revenue, volume, order value, product performance, customer profile and sales trend.

Key metric	Result
Total revenue (sum of Total Amount)	$456,000
Total quantity sold	2,514 units
Number of transactions	1,000
Average order value (AOV)	$456.00
Top category by revenue	Electronics - $156,905 (34.4%)
Customer split	51.0% female / 49.0% male
Average customer age	41.4 years (range 18-64)
Sales trend	Flat with high volatility

- **Revenue is concentrated in high-ticket items.** Products priced at $300 and $500 make up 39.6% of orders but 88.4% of revenue.
- **The three categories perform almost equally.** Electronics leads Clothing by only $1,325 (0.3% of total revenue).
- **Sales are neither growing nor declining meaningfully.** Monthly revenue swings between $23.6K and $53.2K around an average of about $37.9K, and second-half revenue was 4.3% lower than first-half revenue.
- **Every customer bought once.** All 1,000 Customer IDs are unique, so there is no repeat-purchase activity. Retention is the largest untapped growth lever.

2. Objectives and Data

2.1 Business questions

- How much revenue are we generating?
- How many products are we selling?
- What is the average value of each transaction?
- Which product category generates the most revenue?
- Who are our customers (age and gender)?
- Are sales increasing or decreasing over time?

2.2 Dataset description

The dataset contains nine fields per transaction:

Field	Description
Transaction ID	Unique identifier (1 to 1,000)
Date	Transaction date (1 Jan 2023 to 1 Jan 2024)
Customer ID	Unique customer identifier (CUST001 to CUST1000)
Gender	Male or Female
Age	Customer age (18 to 64)
Product Category	Beauty, Clothing or Electronics
Quantity	Units purchased (1 to 4)
Price per Unit	$25, $30, $50, $300 or $500
Total Amount	Quantity x Price per Unit

2.3 Data quality checks

- **Completeness:** no missing values in any field across all 1,000 rows.
- **Uniqueness:** Transaction IDs and Customer IDs are all unique (no duplicates).
- **Consistency:** Quantity x Price per Unit equals Total Amount on every row.
- **Date range:** two transactions fall on 1 January 2024. January 2024 is a partial month, so it is included in overall totals but excluded from monthly trend conclusions.

3. Methodology (Microsoft Excel)

The analysis was carried out in Excel using the following approach.

Metric	Excel approach
Total revenue	=SUM(Total Amount column)
Total quantity	=SUM(Quantity column)
Average order value	=Total revenue / COUNT of Transaction ID (equivalently =AVERAGE of Total Amount)
Revenue by category	PivotTable (Rows: Product Category, Values: Sum of Total Amount) or =SUMIFS
Gender split	PivotTable (Rows: Gender, Values: Count of Customer ID, Sum of Total Amount)
Age groups	Helper column with =IFS(Age<=25,"18-25",Age<=35,"26-35",Age<=45,"36-45",Age<=55,"46-55",TRUE,"56-64"), then PivotTable
Monthly trend	Helper column =TEXT(Date,"yyyy-mm") or PivotTable date grouping by Months, Values: Sum of Total Amount

Age bands are 10-year groups chosen so that each group has a comparable number of customers (169 to 229 orders each). Different band choices may shift group-level results slightly.

4. Findings

4.1 Overall sales performance

The business generated **$456,000** in revenue from **1,000 transactions** and **2,514 units**. The average order value is **$456.00**, or about 2.5 units per order. The median order is only $135, well below the average, because a minority of large orders carries much of the revenue: 201 orders of $1,000 or more produce 63% of total revenue.

4.2 Product category performance

Category	Revenue	Share	Units	Orders	AOV
Electronics	$156,905	34.4%	849	342	$458.79
Clothing	$155,580	34.1%	894	351	$443.25
Beauty	$143,515	31.5%	771	307	$467.48
Total	$456,000	100%	2,514	1,000	$456.00

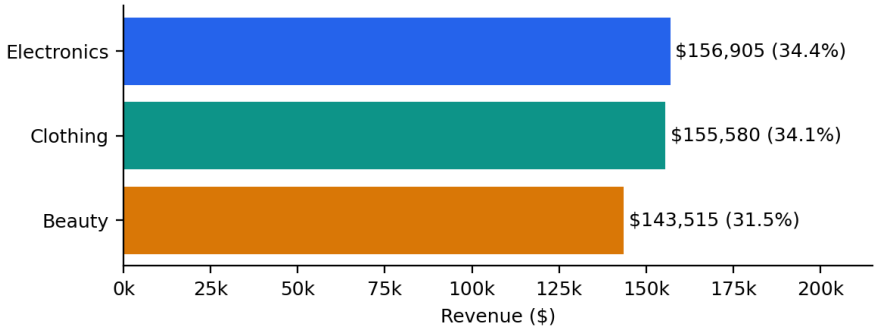


Figure 1. Revenue by product category.

Electronics is the top category by revenue, but the margin is narrow. Clothing sells the most units and has the most orders, while Beauty has the highest AOV but the fewest orders. The practical conclusion is that the three categories are near-equal contributors, so growth depends more on price tier and order volume than on category choice.

By price point, the $500 tier alone produces $247,500 (54.3% of revenue) from 199 orders, and the $300 tier produces $155,400 (34.1%) from 197 orders. The three lowest price points ($25, $30 and $50) represent 60.4% of orders but only 11.6% of revenue.

4.3 Customer demographics

Gender

Gender	Customers	Revenue	Share	AOV	Avg age
Female	510	$232,840	51.1%	$456.55	41.4
Male	490	$223,160	48.9%	$455.43	41.4

The customer base is evenly balanced and spending behavior is almost identical. Female customers lean toward Clothing ($81,275) and Electronics ($76,735), while male customers lean toward Electronics ($80,170).

Age

Age group	Orders	Revenue	Share	AOV
18-25	169	$84,550	18.5%	$500.30
26-35	205	$98,480	21.6%	$480.39
36-45	202	$91,870	20.1%	$454.80
46-55	229	$100,690	22.1%	$439.69
56-64	195	$80,410	17.6%	$412.36

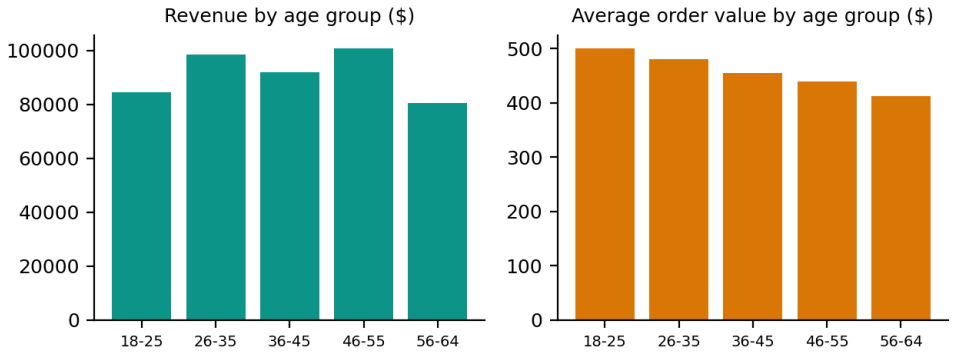


Figure 2. Revenue and average order value by age group.

The 46-55 group generates the most revenue because it has the most customers. The **18-25 group spends the most per order ($500)**, and AOV declines steadily with age to $412 for the 56-64 group, which is also the lowest-revenue segment. Beauty leads among 18-25 customers, Clothing leads among 26-35 customers, and Electronics leads in the three older groups.

4.4 Monthly sales trend

Month (2023)	Revenue	Orders	Units	MoM change
Jan	$35,450	76	195	-
Feb	$44,060	85	214	+24.3%
Mar	$28,990	73	194	-34.2%
Apr	$33,870	86	214	+16.8%
May	$53,150	105	259	+56.9%
Jun	$36,715	77	197	-30.9%
Jul	$35,465	72	176	-3.4%
Aug	$36,960	94	227	+4.2%
Sep	$23,620	65	170	-36.1%
Oct	$46,580	96	252	+97.2%
Nov	$34,920	78	205	-25.0%
Dec	$44,690	91	207	+28.0%

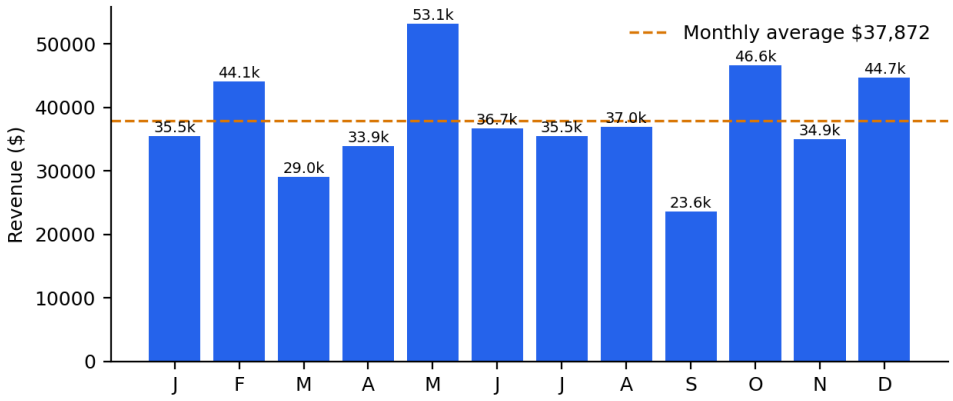


Figure 3. Monthly revenue in 2023 against the monthly average.

Sales are neither clearly increasing nor decreasing. A linear trend line rises by only about $145 per month, which is under 0.4% of the monthly average and is negligible against month-to-month swings of up to $15,000. First-half revenue was $232,235 and second-half revenue was $222,235, a 4.3% decline. By quarter, revenue was Q1 $108,500, Q2 $123,735, Q3 $96,045 and Q4 $126,190.

The peak month was **May ($53,150)** and the lowest was **September ($23,620)**, followed by a 97% rebound in October. These swings are driven mainly by how many $300 and $500 items sell in a month, not by order count alone.

5. Answers to the Business Questions

Question	Answer
How much are we generating?	$456,000 total revenue
How many products are we selling?	2,514 units across 1,000 orders
Average value of each transaction?	$456.00 (median $135)
Top category by revenue?	Electronics, $156,905 (34.4%), narrowly ahead of Clothing and Beauty
Who are our customers?	Evenly split by gender (51% F / 49% M), aged 18-64 with an average of 41; younger customers spend more per order
Are sales rising or falling?	Essentially flat with strong volatility; second half about 4% below first half

6. Dashboard Overview

The accompanying Excel dashboard presents these results visually so that stakeholders can read performance at a glance. It is organised around the six business questions:

- **KPI cards:** total revenue, total quantity sold, average order value and top category.
- **Category chart:** revenue by product category.
- **Demographic charts:** customers and revenue by gender and by age group.
- **Trend chart:** monthly revenue across 2023.

7. Recommendations

1. **Protect and grow the high-ticket mix.** The $300 and $500 items deliver 88% of revenue. Prioritise stock availability, bundling and promotion of these items, and use low-priced items as add-ons rather than expecting them to move revenue.
2. **Treat the three categories as a balanced portfolio.** Beauty has the highest AOV but the lowest volume, so lifting its order count is the cheapest route to growth. Clothing has strong volume, so upselling it to higher price tiers would raise its AOV.
3. **Target by age.** Keep 18-25 customers engaged with Beauty and premium offers. Protect the 46-55 group, the largest revenue source, with Electronics-led loyalty offers. Develop tailored premium bundles for the 56-64 group, which has the lowest AOV.
4. **Smooth the seasonality.** Plan promotions for the weaker months (March, April, July, September, November) and prepare inventory for the peaks (February, May, October, December).
5. **Build customer retention.** All customers purchased exactly once. A repeat-purchase programme such as follow-up emails, loyalty points or personalised category offers would add growth without relying on new-customer acquisition.
6. **Use broad campaigns across genders.** Spending is the same for both genders, so personalise only by category affinity (Clothing for female customers, Electronics for male customers).

8. Limitations

- The data covers a single year, so seasonality cannot be validated against prior years.
- There are no cost or margin figures, so conclusions reflect revenue rather than profit.
- Each customer appears once, so repeat behavior and customer lifetime value cannot be measured.
- Age groups are analyst-defined; alternative bands may shift segment-level results slightly.